WATERWATCH · PROBLEM STATEMENT

WaterWatch problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: users-and-needs and stakeholder tables (Tables 2 and 3) added for the user requirements (WWT-URS-001); problem unchanged
0.4	2026-10-02	Amish Chadha	Partner, alert and data questions answered by Amish's decisions of 2026-10-02 (WWT-DEC-001)
0.3	2026-09-25	Amish Chadha	For TRL 3, record Amish's decisions from WWT-DDR-001 (tapstands first, default thresholds); partner and region left open
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work, open questions)
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWatch problem statement

Rural water points go untested between rare lab visits, so a chlorinator that runs dry or a storm that clouds the source can go unnoticed for weeks while people keep drinking the water.

The problem

In 2024, 2.1 billion people still lacked safely managed drinking water, and about 40 % of rural people had no safely managed service (WHO and UNICEF JMP, 2025; JMP report 2025). Many rural schemes now chlorinate at the tank or with an in-line doser, but the check that the chlorine actually reaches the tap is a manual test with a comparator kit, done when a technician visits.

Two quantities tell an operator most of what they need to act on:

- **Free chlorine residual.** WHO guidance, as summarized for field use by Oxfam, is at least 0.5 mg/L after 30 min of contact at pH below 8, and at least 0.2 mg/L at the point of delivery (Oxfam WASH, chlorination in emergencies). No residual at the tap means the dose is missing or used up, and the water has no protection against recontamination.
- **Turbidity.** Particles shield microbes from chlorine. Chlorination loses effectiveness above about 10 NTU and is not recommended above 100 NTU without pre-treatment (same source); WHO's review of turbidity explains why lower is better (WHO, Water quality and health: review of turbidity).

Remote monitoring has already changed rural water services for **functionality**. Cellular sensors on handpumps in Rwanda shortened repair times when paired with a maintenance service that acted on the data, and similar sensors have tracked functionality and use in Nigeria (Nagel et al., ES&T 2015; Plateau State sensors, 2022; Thomson, WIREs Water 2021). The same is not yet true for **water quality**. Reagent-free in-line chlorine analyzers are built for utilities: an amperometric probe alone lists at about $1,900 (Sensorex FCL) and needs a flow cell, a transmitter and mains power. Low-cost parts exist but are not yet a system: a proven open-source turbidimeter (Kelley et al., Sensors 2014), research graphite chlorine sensors (Pan et al., Anal. Chem. 2015; pencil graphite electrodes, PLOS ONE 2021) and early long-term trials of low-cost chlorine sensors in on-site water reuse (Water Science and Technology, 2025). There is no open, solar-powered reference design that a small operator can build, install on a tapstand and trust enough to act on.

Users and context

Table 1. Users and context (proposed; to be confirmed through co-design).

User	Need	Context
Tapstand caretaker or water committee member	A plain alert when the water at the tap needs attention, and a simple monthly check routine	Village tapstand, kiosk or school tank; phone with SMS, often not a smartphone
Scheme operator or maintenance service provider	Chlorine and turbidity trends from many points, so dosing and repairs can be targeted	Small utility, NGO or private operator serving several villages
District water or health officer	Evidence that treated water reaches people, especially during cholera or flood season	Periodic review; outbreak response
Researcher or open hardware community	A reproducible, inspectable design to build on	University labs, makerspaces, WASH research groups

Table 2. Users and needs (kit 1.8; the source of the user requirements in WWT-URS-001; proposed, to confirm in co-design).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Tapstand caretaker or water committee member	Know when the water at the tap needs attention; a short monthly routine with tools already in the kit	Gets an SMS on a basic phone; opens the cell, wipes it, runs a DPD comparator test and enters the result once a month	No smartphone; little training time; unpaid or part-time; cannot judge a sensor fault from a water problem
Scheme operator or maintenance service provider	Chlorine and turbidity trends from many tapstands; alerts that are trustworthy enough to send a technician	Reviews uploads weekly; acts on alerts by checking the doser or the source	False alarms waste trips; missed events lose trust; budgets of a few hundred dollars per point; parts must be easy to get
District water or health officer	Evidence that treated water reaches people, above all in cholera and flood season	Reads a weekly summary; asks for site data in an outbreak	Data scattered across operators; needs measurements, never a claim that water is safe
Researcher or open hardware community	A reproducible design they can build, test and improve	Builds a unit in a lab or makerspace; compares it with lab methods	Closed commercial analysers cannot be inspected; low-cost sensors lack long-term data
People who drink the water (indirect users)	Water that stays safe; the tapstand works as before	Fill containers at the tap; never touch the unit	A unit that contaminates the supply, takes their water or blocks the tap would harm them

Table 3. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Scheme operator (first candidate: Water Mission, decided by Amish on 2026-10-02, WWT-DEC-001; not yet approached)	Owns the data, installs and maintains units, chooses alert recipients	Co-design partner: sites, thresholds, alert wording, maintenance routine
Water committee and community	Hosts the unit at its tapstand; agrees where flushed water goes	Consent for installation; caretaker nomination; drain location
District water and health offices	Use the data; may set rules for monitoring	Agree the weekly summary and outbreak access
National regulators (water quality, telecommunications, battery transport)	Drinking water standards, radio type approval, lithium battery shipping	Checked in the regulatory register (WWT-REG-001); type approval before any field unit
Mobile network operators	Carry uploads and SMS	Coverage and SIM plans per site
Open hardware community and funders	Build, review and fund the open design	Repository, build reports, citations

Typical site: a public tapstand or a storage tank outlet on a piped rural scheme with chlorination at the source or tank. No mains power. Ambient temperature from about 0 to 45 °C, direct sun, dust and splashing. Cellular coverage is usually 2G or LTE-M/NB-IoT; some districts have a LoRaWAN gateway.

Constraints

- Garage-buildable prototype, about $300 USD in parts per unit (`project.yaml`), using off-the-shelf modules and simple fabricated parts.
- Solar powered, with no mains connection and no change to the tapstand beyond one tee on the riser.
- Must not waste much treated water, and must not contaminate the supply (backflow and material safety).
- Must be maintainable by a trained caretaker with a comparator kit, without special tools.
- Must not claim that water is "safe". It reports measurements and alerts; accredited lab testing, including microbiological testing, stays necessary.

Out of scope

- Microbiological detection (for example *E. coli*); the sensor watches indicators only.
- Chemical contaminants such as arsenic, fluoride or nitrate.
- Controlling the chlorine dose; WaterWatch observes and alerts, it does not dose.
- Household point-of-use treatment.

Co-design checklist

- [ ] Identify a partner scheme operator or maintenance service and two or three pilot sites.
- [ ] Interview caretakers, operators and district officers on alert content, language, thresholds and who acts.
- [ ] Confirm local cellular coverage and SMS costs, or LoRaWAN availability.

- [ ] Agree where flushed sample water goes at each site.
- [ ] Agree who owns the data and who may see it.

Open questions

- Which partner and region first? Decided by Amish, 2026-10-02 (WWT-DEC-001): a scheme operator that runs chlorinated, solar-powered piped schemes with public tapstands; the first candidate to approach is Water Mission, which runs such schemes in East Africa, with the first region taken from its country programs; not yet approached.
- Tapstand only, or also storage tank outlets and kiosks? Decided by Amish, 2026-09-25: tapstands first, tank outlets and kiosks later (WWT-DDR-001, D8).
- Which alert thresholds should be defaults? Decided by Amish, 2026-09-25: free chlorine below 0.2 mg/L and turbidity above 5 NTU, confirmed by two consecutive readings and adjustable per site (WWT-DDR-001, D6). Partners may still ask for other site values.
- Alert language, recipients and data ownership: decided by Amish, 2026-10-02 (WWT-DEC-001) as defaults to confirm in co-design: alerts go by SMS in the site's main local language, with an English copy to the operator, stating only what was measured, when and the threshold crossed; they go to the caretaker and the operator's maintenance contact, with a weekly summary to the district water officer; the scheme operator owns the data; the water committee and the district water and health offices can see it; anonymized site data are published only with the operator's written agreement.